CentraleSupélec

DECISION MODELLING
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PROJECT
Staffing collaborators on projects

1 Context

We are focusing on the company *CompuOpti*, which develops and implements optimization and decision-support solutions for its clients. *CompuOpti* employs a number of software engineers who are assigned to client projects. Each project requires a specific number of person-days involving particular skill sets (optimization, project management, web development, etc.). For instance, a project might require 6 person-days of skill A, 2 person-days of skill B, and 5 person-days of skill C. Staff allocation and scheduling are critical issues for *CompuOpti*, and you have been commissioned to develop a solution for Margaux Dourtille, the company's CEO, to enable her to efficiently schedule staff for projects (bearing in mind that not all projects can always be staffed).

The problem under study is therefore one of staff scheduling and project allocation. We assume the problem takes place over a specific time horizon (considering only working days). Each *CompuOpti* staff member possesses specific qualifications drawn from a predefined set (e.g., $\{A, B, C, D, E\}$) and has scheduled leave days falling within the time horizon.

Each project requires specific qualifications from the available set (a subset of $\{A, B, C, D, E\}$). Each qualification involved in a project is associated with a specific number of workdays dedicated to that skill. Furthermore, each project generates revenue if completed, and Margaux aims to maximize the total profit derived from the completed projects.

The task, therefore, is to define schedules for staff members—that is, to assign each staff member's workday to a specific project role (or to no activity). The figure below shows a simple scheduling example involving 5 people (Alice, Bob, Charlie, David, and Eve), 3 skills (A, B, and C), and 5 projects (I, II, III, IV, and V).

eve	II,C	II,C	II,C	II,C	III,C	III,C	III,C	IV,C	V,C	V,C	V,C	
david	I,B	I,B	I,B	II,B	II,B	IV,B	IV,B	IV,B				
charlie	I,A	I,A	II,A	II,A	II,A	III,A	IV,A	IV,A	IV,A			
bob	I,C	I,C	II,C	II,C	II,C	III,C	III,C	IV,C	IV,C	V,C		
alice		I,A	I,A	III,A	III,A		III,A	III,B		V,A	V,B	
	1	2	3	4	5	6	7	8	9	10	11	12

Personnel	
nom	compétences
alice	A,B
bob	C
charlie	A
david	B
eve	A,B,C

	Projets		
	A	B	C
I	4	3	2
II	3	2	7
III	4	1	5
IV	3	3	3
V	1	1	4

When creating the schedule, a number of constraints must naturally be respected:

- A staff member can only be assigned to a project qualification if they possess that qualification (staff qualification constraint).
- At any given time, an employee can be assigned to only one project and one specific qualification within that project (daily staff assignment uniqueness constraint).
- A staff member cannot be assigned to a project qualification on a day off (leave constraint).
- A project is considered completed only if all workdays dedicated to each of the project's qualifications have been covered by staff members (project qualification coverage constraint).
- Finally, a project can be executed only once within a given time period (project execution uniqueness constraint).

To identify suitable schedules, it may be reasonable to consider the following criteria:

- Maximize total profit realized,
- Minimize the actual project execution time,
- Minimize the number of projects an employee works on,
- Minimize the number of projects delivered late,
- ...

2 Datasets

To test your model, you will have access to three instances of increasing size (`toy_instance.json`, `medium_instance.json`, and `large_instance.json`) in JSON format, available for download from the course's Edunao page. In addition to these three initial instances, you will need to build an instance generator to test your algorithms' performance.

3 Organization and Tasks

The project is to be carried out in groups of three. Each group will present their project during a defense.

The project aims to calculate the set of the non-dominated solutions for the multi-objective optimization problem you formulate. Your code must allow for the visualization of non-dominated solutions in objective space, as well as the individual solutions in decision space. For the final deliverable (to be submitted at the final defense), you must:

- update your Git repository.
- Prepare slides containing a precise description of the models you worked on, as well as the results obtained by the different models on the provided datasets and any additional datasets you created.

Some tips for mathematical modeling.

- List the problem parameters (e.g., time horizon, set of qualifications, etc.) required for the model and assign them compact symbols (e.g., h for the time horizon and $\mathcal{Q}$ for qualifications). Note that these parameters can be integers (like h) or sets (like $\mathcal{Q}$).
- Define decision variables and assign them compact symbols (ideally a single letter, e.g., X). Also, provide an interpretation for them (e.g., $X = 1$ if..., 0 otherwise). Do not forget to specify the set of values these variables can take (e.g., $\{0, 1\}$). If the variables involve indices, be sure to precisely define the range of values those indices can take.
  Note: Be careful to distinguish between problem parameters—which are fixed data—and decision variables, which represent choices.
- Define the problem constraints. As with decision variables, provide an interpretation for them; if they involve indices, remember to specify the range of values those indices can take.
- Define the problem objectives.
- Finally, ensure there are no notation conflicts. In other words, avoid using the same symbol to represent different things.
- If your formulation leads to an exact solution method with high computational cost, it may be wise to forgo optimality and consider an approximate solution instead.